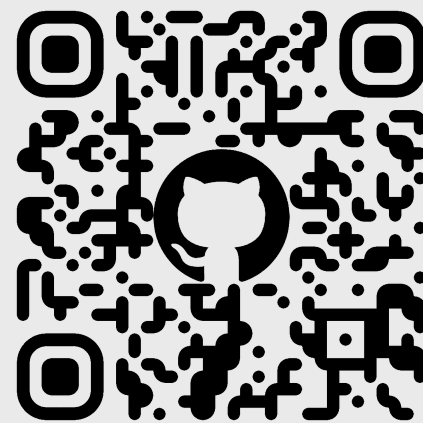


Replacing Convolution with Kolmogorov–Arnold Operators

An Empirical Analysis of Expressivity, Parameter Efficiency, and Training Cost

AUTHOR AFFILIATIONS

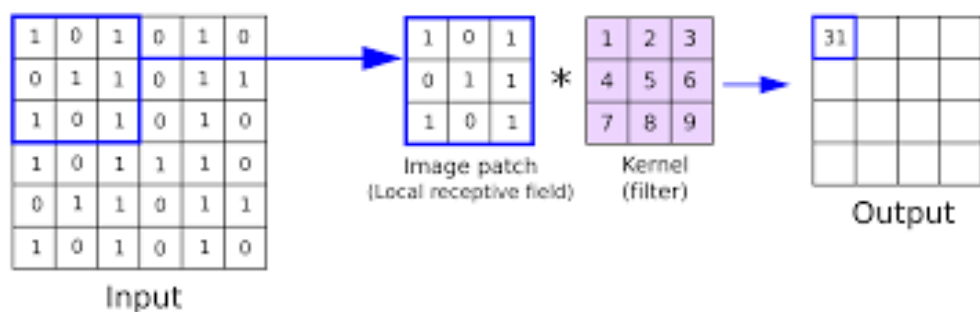
Dejan Lisica University of Novi Sad



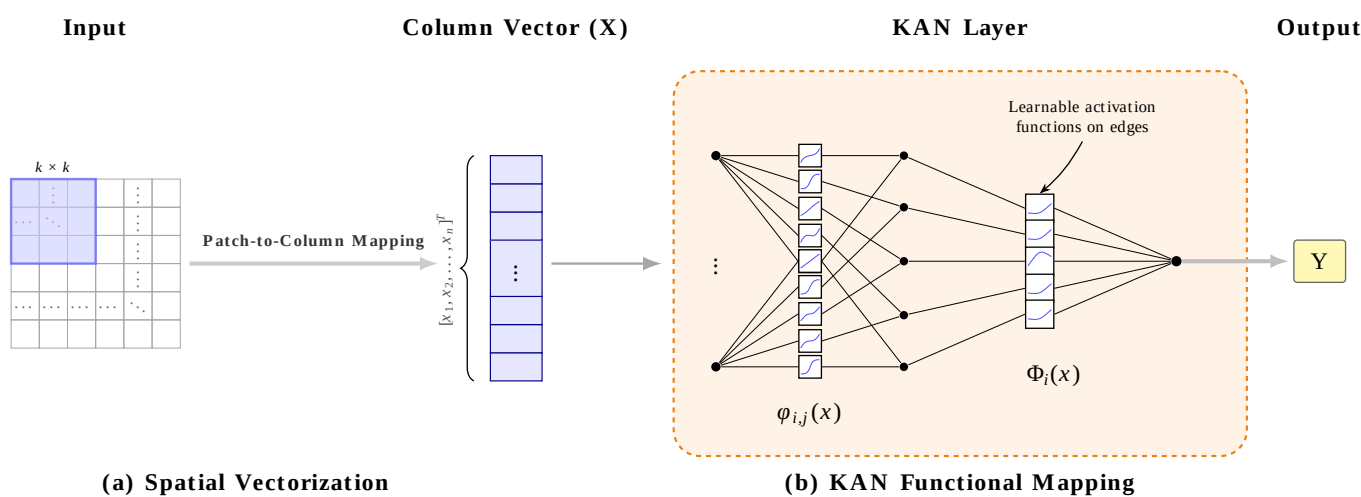
MOTIVATION

- Convolutions are linear local operators followed by nonlinear activation.
- Kolmogorov–Arnold Networks (KANs) model nonlinear functions directly via spline bases.
- Hypothesis: replacing linear kernels with nonlinear spline operators may increase local expressivity.

STANDARD CONVOLUTION



PROPOSED OPERATOR



TINY MODEL STUDY

ARCHITECTUAL BACKBONE

TINY-CONVNET (BASELINE)

- 3x3 CONV → BN → RELU
- DOWNSAMPLING VIA STRIDE 2
- GLOBAL AVERAGE POOLING
- LINEAR CLASSIFIER

TINY-KANNET (PROPOSED)

- SAME BACKBONE AS TINY-CONVNET
- ADDITIONAL 3x3 KAN OPERATOR
- SAME DEPTH AND CHANNEL CONFIGURATION
- SAME POOLING AND CLASSIFIER

EXPERIMENT

- DATASETS: CIFAR-10, CIFAR-100
- IMAGE SIZE: 32x32
- TRAINING: 100 EPOCHS
- OPTIMIZER: ADAM
- SEEDS: 0,1,2
- SAME TRAINING PIPELINE FOR BOTH MODELS

Results

Model	CIFAR-10	CIFAR-100	Parameters
Tiny Conv	61.95 ± 0.33%	32.02 ± 0.34%	~95k
Tiny KAN	70.58 ± 0.43%	40.52 ± 0.30%	~1.58M

- EVALUATION METRIC: TOP-1 TEST ACCURACY (%)

FINDINGS

- TinyKANNet achieves higher mean accuracy on CIFAR-10 and CIFAR-100.
- Improvement is consistent across 3 seeds.
- Parameter count increases significantly (~16x).
- Training time per epoch approximately doubles.

INTERPRETATION

- Adding a nonlinear KAN spatial operator at low resolution improves classification accuracy on CIFAR-10 and CIFAR-100.
- The improvement is consistent across 3 random seeds.
- Gains come with significantly increased parameter count and training time.
- Results suggest KAN operators can increase representational capacity, but are not currently parameter- or compute-efficient.

ARCHITECTURAL MODIFICATION (RESNET18 → KAN-RESNET18-TINY)

BACKBONE MODIFICATION

We construct KAN-ResNet18-Tiny by:

- Preserving ResNet18 depth (2-2-2 blocks)
- Keeping residual connections and classifier unchanged
- Replacing 3x3 convolutions in Layer4 only with KAN operators
- Using width multiplier 0.5 (Tiny variant)

EXPERIMENT

- DATASETS: CIFAR-10, CIFAR-100, OXFORD-IIIT PETS, FOOD101
- TRAINING: 100 EPOCHS
- OPTIMIZER: ADAM
- SEEDS: 0
- SAME TRAINING PIPELINE FOR ALL MODELS

CIFAR (32x32)			
Model	Parameters	CIFAR-10	CIFAR-100
KAN-ResNet18-Tiny	21.7M	0.44	0.675
ResNet18	11.2M	0.728	0.442
ResNet50	23.5M	0.66	0.387
VGG16	134.3M	0.796	0.479
MobileNetV2	2.2M	0.64	0.35
ShuffleNetV2	1.3M	0.632	0.355
EfficientNet-B0	4M	0.681	0.395
EfficientNet-B1	6.5M	0.641	0.342

High-Resolution Benchmarks (224x224)			
Model	Parameters	Pets	Food101
KAN-ResNet18-Tiny	21.7M	0.44	0.675
ResNet18	11.2M	0.436	0.706
ResNet50	23.7M	0.28	0.726
VGG16	134.4M	0.175	0.712
MobileNetV2	2.3M	0.372	0.708
ShuffleNetV2	1.3M	0.343	0.678
EfficientNet-B0	4.1M	0.441	0.709
EfficientNet-B1	6.6M	0.413	0.704

FINDINGS

- KAN-ResNet18-Tiny achieves performance comparable to standard ResNet18 across datasets.
- It does not provide consistent accuracy improvements at backbone scale.
- The model requires ~2x more parameters than ResNet18.
- Training time is higher than the standard convolutional backbone.

INTERPRETATION

- Increasing local kernel expressivity via nonlinear KAN operators does not systematically improve deep residual architectures.
- The additional parameter and computational cost does not translate into proportional performance gains.
- The benefit of nonlinear spline-based kernels appears limited when integrated into large-scale CNN backbones.